LEARNING HUB MASTER SERIES • VOLUME 05

Linear Transformations & Change of Basis

Axiomatic Mappings, Operator Geometry in $\mathbb{R}^2$ & $\mathbb{R}^3$, Matrix Representation of Linear Operators, Coordinate Transformations, Similarity, and the Canonical Search for Eigenbases.

Master Reference Scope & Foundations:

Covers the complete algebraic and geometric theory of linear transformations: axiomatic foundations, kernel and range spaces, the Dimension Theorem (Rank-Nullity), the systematic 3-step matrix construction recipe, differential operators on function spaces, basis transformation matrices M, the similarity transformation law $B = M^{-1}AM$, diagonalization via eigenbases, and optimal singular vector bases for rectangular operators.

Curriculum & Pedagogical Alignment:

- Gilbert Strang — *Introduction to Linear Algebra* (5th Edition, Chapter 8)
- Kenji Hiranabe — *The Art of Linear Algebra* (Matrix World & Visual Decompositions)
- Learning Hub Core Mathematics Curriculum — *Topic: Linear Transformations*

Publication Standard: Pure English Master Edition • Document/Linear-Algebra/ Directory

Features: 4 Scalable Vector Diagrams • 4 Amber ⚡ Fast-Tracks • 5 Step-by-Step Worked Problems • Matrix Proofs

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Part I: The Idea of a Linear Transformation

Chapter 1: Axiomatic Definition, Mapping Linearity & Counterexamples

In classical matrix algebra, an $m \times n$ matrix A multiplies a column vector $\mathbf{x} \in \mathbb{R}^n$ to produce an output vector $A\mathbf{x} \in \mathbb{R}^m$. However, linear algebra reaches its full generality and expressive power when viewed *coordinate-free*: as abstract vector spaces acted upon by operations that preserve their internal algebraic structure.

▶ Definition 1.1: Linear Transformation (Axiomatic Definition)

Let V and W be real vector spaces. A function $T: V \to W$ is called a **linear transformation** (or **linear operator** when $W = V$) if and only if it satisfies two governing axioms for all vectors $\mathbf{v}, \mathbf{w} \in V$ and all real scalars $c, d \in \mathbb{R}$:

1. **Additivity:** $T(\mathbf{v} + \mathbf{w}) = T(\mathbf{v}) + T(\mathbf{w})$
2. **Homogeneity (Scalar Multiplication):** $T(c\mathbf{v}) = c\,T(\mathbf{v})$

Equivalently, both axioms combine into a single, unified linearity condition:

$$T(c\mathbf{v} + d\mathbf{w}) = c\,T(\mathbf{v}) + d\,T(\mathbf{w}) \qquad \forall\, \mathbf{v}, \mathbf{w} \in V, \quad c, d \in \mathbb{R}$$

1.1 Essential Algebraic Consequences of Linearity

The two linearity axioms directly impose strict structural constraints on how T reshapes the domain:

- **The Origin is Fixed:** Every linear transformation maps the domain zero vector strictly to the codomain zero vector:

  $$T(\mathbf{0}_V) = \mathbf{0}_W$$

  Proof: Set scalar $c = 0$ in the homogeneity axiom: $T(\mathbf{0}_V) = T(0 \cdot \mathbf{v}) = 0 \cdot T(\mathbf{v}) = \mathbf{0}_W$. ■

- **Preservation of All Linear Combinations:** By mathematical induction, linearity extends over any finite superposition of k vectors:

  $$T\left(\sum_{i=1}^{k} c_i \mathbf{v}_i\right) = \sum_{i=1}^{k} c_i\, T(\mathbf{v}_i)$$

- **Straight Lines Map to Straight Lines (or Points):** Consider a parametric line segment joining $\mathbf{u}$ and $\mathbf{v}$ in V: $\mathbf{x}(t) = (1 - t)\mathbf{u} + t\mathbf{v}$ for $t \in [0, 1]$. Applying T yields:

  $$T(\mathbf{x}(t)) = (1 - t)T(\mathbf{u}) + tT(\mathbf{v})$$

  This is precisely the equation of a line segment connecting $T(\mathbf{u})$ and $T(\mathbf{v})$ in W. If $T(\mathbf{u}) = T(\mathbf{v})$, the entire line collapses to a single point. Under no circumstances can a linear transformation bend a straight line into a curve.

1.2 Linear vs Non-Linear Transformations: Concrete Counterexamples

Many common geometric operations and algebraic formulas fail the linearity test. A thorough understanding of counterexamples is crucial:

Mapping Candidate	Definition & Action	Linear?	Axiomatic Failure Mechanism
Affine Shift / Translation	$T(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$ $(\mathbf{b} \neq \mathbf{0})$	NON-LINEAR	$T(\mathbf{0}) = \mathbf{b} \neq \mathbf{0}$; also $T(\mathbf{u}+\mathbf{v}) = A\mathbf{u}+A\mathbf{v}+\mathbf{b} \neq T(\mathbf{u})+T(\mathbf{v})$
Length / Norm Map	$T(\mathbf{x}) = \lVert\mathbf{x}\rVert$ (from $\mathbb{R}^n \to \mathbb{R}$)	NON-LINEAR	Fails homogeneity for negative scalars: $T(-\mathbf{x}) = \lVert-\mathbf{x}\rVert = \lVert\mathbf{x}\rVert \neq -T(\mathbf{x})$
Quadratic / Cross-Product	$T(x_1, x_2) = x_1 x_2$	NON-LINEAR	$T(c\mathbf{x}) = (cx_1)(cx_2) = c^2 x_1 x_2 = c^2 T(\mathbf{x}) \neq cT(\mathbf{x})$
Trigonometric Distortion	$T(x) = \sin(x)$	NON-LINEAR	$\sin(u + v) = \sin u \cos v + \cos u \sin v \neq \sin u + \sin v$
Matrix Multiplication	$T(\mathbf{x}) = A\mathbf{x}$ ($m \times n$ matrix)	LINEAR	$A(c\mathbf{v} + d\mathbf{w}) = c(A\mathbf{v}) + d(A\mathbf{w})$ satisfies both axioms perfectly
Derivative Operator	$T(f) = \frac{df}{dx}$ on $\mathscr{C}^1(\mathbb{R})$	LINEAR	$\frac{d}{dx}(cf + dg) = c\frac{df}{dx} + d\frac{dg}{dx}$ by calculus rules

⚡ FAST-TRACK 1: Quick Linearity & 2D Matrix Inspection Mnemonic

To test linearity in 3 seconds:

1. **Step 1 (The Zero Check):** Plug in $\mathbf{0}$. If $T(\mathbf{0}) \neq \mathbf{0}$, immediately reject: *it is not linear*! (Catches all affine translations like $x + 3$).
2. **Step 2 (The Degree Check):** Look at the formulas for the output coordinates. Every single term must be strictly degree-1 in the input variables (no squares, square roots, cross-products xy, or constants).
3. **Step 3 (Immediate 2D Matrix Assembly):** For any linear map on $\mathbb{R}^2$, the matrix A is written down instantly by tracking where the standard unit vectors land:

$$A = \begin{vmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) \end{vmatrix} = \begin{vmatrix} T\begin{vmatrix}1\\0\end{vmatrix} & T\begin{vmatrix}0\\1\end{vmatrix} \end{vmatrix}$$

The destination of $[1, 0]^T$ forms Column 1; the destination of $[0, 1]^T$ forms Column 2. No algebra required!

Part II: Geometric Transformations in $\mathbb{R}^2$ and $\mathbb{R}^3$

Chapter 2: The Canonical 2D Operator Gallery

Every linear transformation T: $\mathbb{R}^2 \to \mathbb{R}^2$ can be completely comprehended by watching its effect on the unit square spanned by $\mathbf{e}_1 = [1, 0]^T$ and $\mathbf{e}_2 = [0, 1]^T$. Because T is linear, the unit square always transforms into a parallelogram whose sides are the column vectors $T(\mathbf{e}_1)$ and $T(\mathbf{e}_2)$.

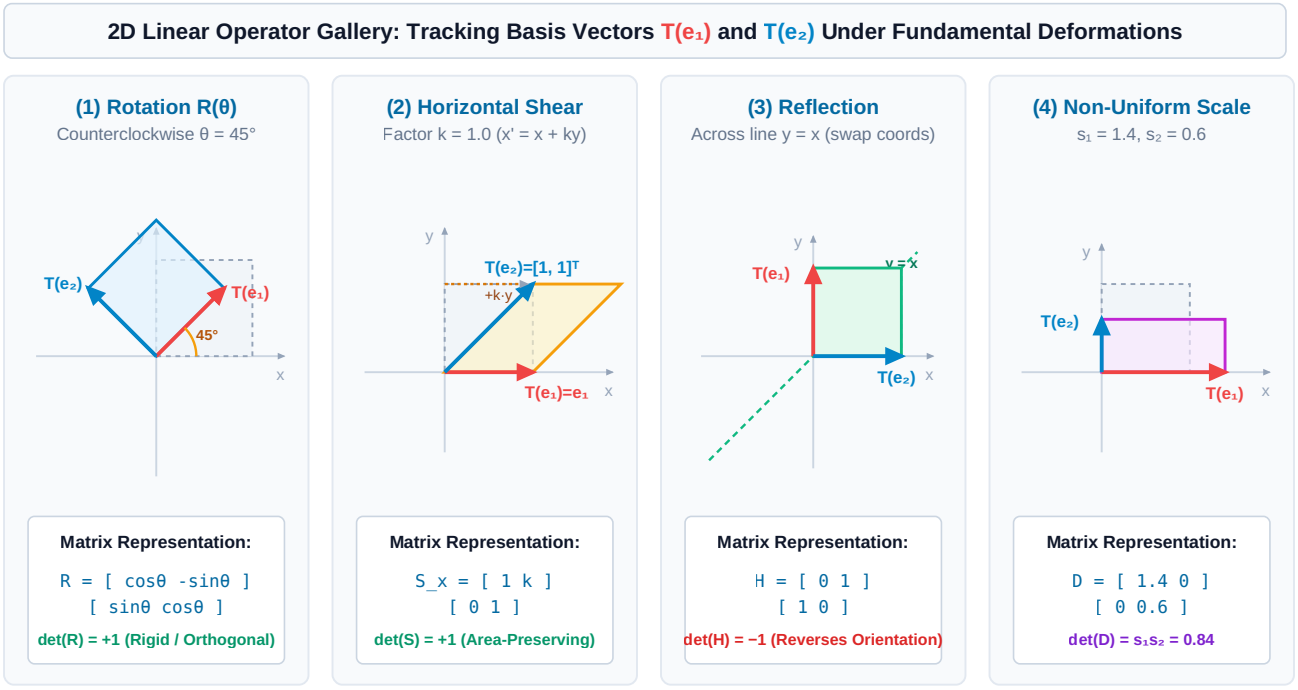


Figure 5.2: The 2D Linear Transformation Gallery illustrating the action of Rotation, Shear, Reflection, and Scaling on basis vectors $\mathbf{e}_1$ and $\mathbf{e}_2$.

2.1 Detailed Formulation of the Five 2D Archetypes

1. Counterclockwise Rotation by Angle θ:

The unit vector $\mathbf{e}_1 = [1, 0]^T$ rotates to $[\cos\theta, \sin\theta]^T$, while $\mathbf{e}_2 = [0, 1]^T$ rotates to $[-\sin\theta, \cos\theta]^T$. Placing these as columns gives the orthogonal rotation matrix:

$$R_\theta = \begin{vmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{vmatrix}, \quad \det(R_\theta) = \cos^2\theta + \sin^2\theta = +1, \quad R_\theta^{-1} = R_{-\theta} = R_\theta^T$$

Because $\det(R) = +1$ and $R^T R = I$, rotations preserve lengths, angles, areas, and orientation.

2. Reflection Across a Line at Angle α Through the Origin:

Reflecting across a mirror line tilted at angle α to the x-axis maps a vector at angle θ to $2\alpha - \theta$. The standard unit vector $\mathbf{e}_1$ (at $\theta = 0$) lands at angle 2α, giving $[\cos 2\alpha, \sin 2\alpha]^T$. The unit vector $\mathbf{e}_2$ (at $\theta = \pi/2$) lands at $2\alpha - \pi/2$, giving $[\sin 2\alpha, -\cos 2\alpha]^T$:

$$H_{2\alpha} = \begin{vmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{vmatrix}, \quad \det(H) = -\cos^2 2\alpha - \sin^2 2\alpha = -1, \quad H^2 = I \quad \text{(Involutory)}$$

Alternatively, using a unit normal vector **n** perpendicular to the mirror line, the reflection is given by the Householder formula: $H = I - 2\mathbf{n}\mathbf{n}^T$. The determinant is −1, indicating that reflection reverses geometric orientation (handedness).

3. Orthogonal Projection Onto a Subspace:

Projecting onto a line spanned by unit direction vector $\mathbf{a} = [a_1, a_2]^T$ ($\|\mathbf{a}\| = 1$) yields:

$$P = \mathbf{a}\mathbf{a}^T = \begin{vmatrix} a_1^2 & a_1 a_2 \\ a_1 a_2 & a_2^2 \end{vmatrix}, \quad P^2 = P \quad \text{(Idempotent)}, \quad P^T = P \quad \text{(Symmetric)}$$

Because projecting twice is identical to projecting once ($P^2 = P$), its eigenvalues are strictly $\lambda \in \{0, 1\}$. Since $\det(P) = 0$, projections crush full 2D space down to a 1D line, losing one degree of freedom.

4. Horizontal and Vertical Shears:

A horizontal shear shifts each point horizontally proportional to its height: $x_{new} = x + ky$, while $y_{new} = y$. The base $\mathbf{e}_1$ stays fixed while $\mathbf{e}_2$ slants to $[k, 1]^T$:

$$S_x(k) = \begin{vmatrix} 1 & k \\ 0 & 1 \end{vmatrix}, \quad S_y(k) = \begin{vmatrix} 1 & 0 \\ k & 1 \end{vmatrix}, \quad \det(S) = 1 \quad \text{(Strictly Area-Preserving)}$$

5. Scaling (Uniform vs Non-Uniform):

Uniform scaling by factor s multiplies every coordinate equally: $D = sI$ ($\det = s^2$). Non-uniform scaling stretches axes independently by factors s_1 and s_2:

$$D(s_1, s_2) = \begin{vmatrix} s_1 & 0 \\ 0 & s_2 \end{vmatrix}, \quad \det(D) = s_1 s_2 \quad \text{(Area scaling factor)}$$

Chapter 3: 3D Transformations & Rotations

In $\mathbb{R}^3$, transformations act on volume elements. The determinant $\det(A)$ measures the signed volume change of the unit cube.

- **Euler Axis Rotations:** Rotating counterclockwise around the standard coordinate axes:

$$R_z(\theta) = \begin{vmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix}, \quad R_y(\theta) = \begin{vmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{vmatrix}, \quad R_x(\theta) = \begin{vmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{vmatrix}$$

- **Rodrigues' Rotation Formula:** Rotating by angle θ around an arbitrary unit axis $\mathbf{n} = [n_1, n_2, n_3]^T$:

$$R(\mathbf{n}, \theta) = I + (\sin\theta)\,K + (1 - \cos\theta)\,K^2$$

where K is the skew-symmetric cross-product matrix: $K = \begin{vmatrix} 0 & -n_3 & n_2 \\ n_3 & 0 & -n_1 \\ -n_2 & n_1 & 0 \end{vmatrix}$.

- **3D Reflection Across a Plane:** For a plane with unit normal vector **n** passing through the origin:

$$H_{\text{plane}} = I - 2\mathbf{n}\mathbf{n}^{\mathsf{T}}, \qquad \det(H_{\text{plane}}) = -1$$

Part III: Kernel, Range, and the Dimension Theorem

Chapter 4: Kernel (Nullspace) and Range (Column Space)

Every linear transformation partitions its domain and codomain into two fundamental subspaces that mirror the nullspace and column space of a matrix.

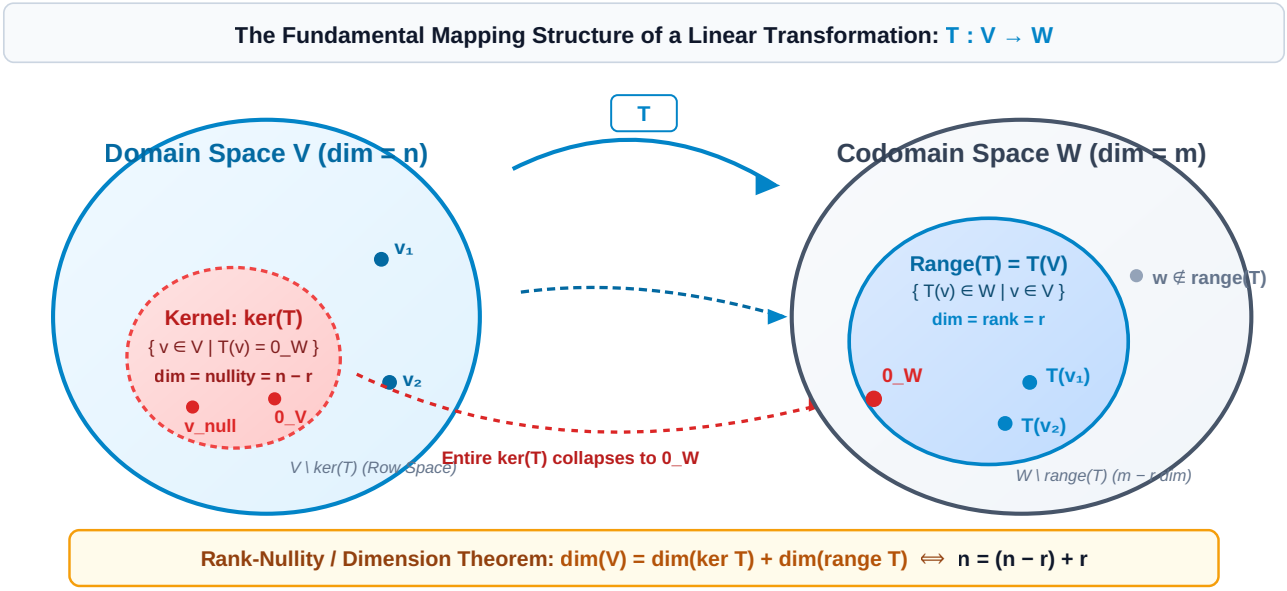


Figure 5.1: The Fundamental Mapping Structure of a Linear Transformation $T: V \to W$, illustrating the collapse of ker(T) to $\mathbf{0}_W$ and the range subspace within W.

▶ Definition 4.1: Kernel and Range Subspaces

Let $T: V \to W$ be a linear transformation.

1. The **Kernel** (or *Nullspace*) of T, denoted ker(T) or $N(T)$, is the set of all vectors in V that map to $\mathbf{0}_W$:

$$\ker(T) = \{ \mathbf{v} \in V \mid T(\mathbf{v}) = \mathbf{0}_W \} \subseteq V$$

2. The **Range** (or *Image*) of T, denoted range(T) or $T(V)$, is the set of all image vectors produced in W:

$$\text{range}(T) = \{ \mathbf{w} \in W \mid \mathbf{w} = T(\mathbf{v}) \text{ for some } \mathbf{v} \in V \} \subseteq W$$

The dimension of the range is the **rank** of T: r = dim(range T).
The dimension of the kernel is the **nullity** of T: nullity = dim(ker T).

🔍 **Proof: Kernel and Range are Genuine Subspaces**

To prove ker(T) is a subspace of V:

1. *Zero vector:* $T(\mathbf{0}_V) = \mathbf{0}_W$ &implies; $\mathbf{0}_V \in \ker(T)$.
2. *Closure under linear combinations:* If $\mathbf{u}, \mathbf{v} \in \ker(T)$, then $T(\mathbf{u}) = \mathbf{0}$ and $T(\mathbf{v}) = \mathbf{0}$. For any scalars c, d, linearity yields: $T(c\mathbf{u} + d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v}) = c\mathbf{0} + d\mathbf{0} = \mathbf{0}$. Thus $c\mathbf{u} + d\mathbf{v} \in \ker(T)$. Hence, ker($T$) is a subspace of V.

To prove range(T) is a subspace of W:

1. *Zero vector:* $\mathbf{0}_W = T(\mathbf{0}_V) \in \text{range}(T)$.
2. *Closure:* If $\mathbf{w}_1, \mathbf{w}_2 \in \text{range}(T)$, there exist $\mathbf{v}_1, \mathbf{v}_2 \in V$ such that $T(\mathbf{v}_1) = \mathbf{w}_1$ and $T(\mathbf{v}_2) = \mathbf{w}_2$. Then $c\mathbf{w}_1 + d\mathbf{w}_2 = cT(\mathbf{v}_1) + dT(\mathbf{v}_2) = T(c\mathbf{v}_1 + d\mathbf{v}_2)$. Since $c\mathbf{v}_1 + d\mathbf{v}_2 \in V$, its image is in range($T$). ■

4.1 Invertibility, Injectivity, and Surjectivity

- **One-to-One (Injective):** T is injective if and only if $\ker(T) = \{\mathbf{0}_V\}$ (nullity = 0). *Proof:* If $T(\mathbf{u}) = T(\mathbf{v})$, then $T(\mathbf{u} - \mathbf{v}) = \mathbf{0}$. If $\ker(T) = \{\mathbf{0}\}$, then $\mathbf{u} - \mathbf{v} = \mathbf{0}$ &implies; $\mathbf{u} = \mathbf{v}$.
- **Onto (Surjective):** T is surjective if and only if range(T) = W (rank = dim W).
- **Isomorphism (Bijective):** T is an isomorphism if it is both injective and surjective. In this case, dim V = dim W, and an inverse linear transformation $T^{-1}: W \to V$ exists uniquely.

Chapter 5: The Dimension Theorem (Rank-Nullity) & Rigorous Proof

▶ **Theorem 5.1: The Dimension Theorem (Rank-Nullity Theorem)**

Let V be a finite-dimensional vector space, and let $T: V \to W$ be any linear transformation. Then:

$$\dim(\ker T) + \dim(\text{range } T) = \dim V$$

In terms of matrix parameters: **Nullity + Rank = *n*** (where n is the dimension of the domain V).

🔍 Complete Step-by-Step Proof of the Dimension Theorem

Let dim $V = n$, and let dim(ker T) = $k \le n$.

1. Choose a basis for the subspace ker(T): $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\}$.
2. By the Basis Extension Theorem of linear algebra, extend this linearly independent set to form a complete basis for V: $\{\mathbf{u}_1, \ldots, \mathbf{u}_k, \mathbf{v}_1, \ldots, \mathbf{v}_r\}$, where $k + r = n$.
3. We now claim that the set of image vectors $S = \{T(\mathbf{v}_1), T(\mathbf{v}_2), \ldots, T(\mathbf{v}_r)\}$ forms a basis for range(T). If this claim holds, then dim(range T) = r, which directly proves $k + r = n$.
4. **Step A (Spanning):** Let $\mathbf{w} \in$ range(T). Then $\mathbf{w} = T(\mathbf{x})$ for some $\mathbf{x} \in V$. Expand $\mathbf{x}$ in the basis of V: $\mathbf{x} = \sum_{i=1}^{k} a_i \mathbf{u}_i + \sum_{j=1}^{r} b_j \mathbf{v}_j$. Applying T:

$$\mathbf{w} = T(\mathbf{x}) = \sum_{i=1}^{k} a_i T(\mathbf{u}_i) + \sum_{j=1}^{r} b_j T(\mathbf{v}_j)$$

   Since each $\mathbf{u}_i \in \ker(T)$, $T(\mathbf{u}_i) = \mathbf{0}$. Thus $\mathbf{w} = \sum_{j=1}^{r} b_j T(\mathbf{v}_j)$. Therefore, S spans range(T).
5. **Step B (Linear Independence):** Suppose a linear combination of S equals zero: $c_1 T(\mathbf{v}_1) + \ldots + c_r T(\mathbf{v}_r) = \mathbf{0}_W$. By linearity: $T(c_1\mathbf{v}_1 + \ldots + c_r\mathbf{v}_r) = \mathbf{0}_W$. This means the vector $\mathbf{v}^* = \sum_{j=1}^{r} c_j \mathbf{v}_j$ lies in ker(T)! Because $\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ spans ker(T), we can write $\mathbf{v}^* = \sum_{i=1}^{k} d_i \mathbf{u}_i$. Equating both expressions:

$$c_1 \mathbf{v}_1 + \ldots + c_r \mathbf{v}_r - d_1 \mathbf{u}_1 - \ldots - d_k \mathbf{u}_k = \mathbf{0}$$

   Since $\{\mathbf{u}_1, \ldots, \mathbf{u}_k, \mathbf{v}_1, \ldots, \mathbf{v}_r\}$ is a basis for V, all coefficients must vanish: $c_1 = \ldots = c_r = 0$ (and all $d_i = 0$). Hence, S is linearly independent!

Having shown that S spans range(T) and is linearly independent, S is a basis for range(T). Therefore, dim(range T) = r. Since $k + r = n$, we conclude: dim(ker T) + dim(range T) = dim V. ■

⚡ FAST-TRACK 2: Rank-Nullity Dimension Verification & Conservation Law

The Conservation of Input Dimensions:

[Total Input Degrees of Freedom] = [Lost / Crushed Dimensions] + [Surviving Output Dimensions]

- **Domain Dimension Governs:** The sum is ALWAYS equal to the dimension of the **Domain** V, NEVER the codomain W!
- **Sanity Check on Projections:** For projection onto a plane in $\mathbb{R}^3$, the output is a 2D plane (rank = 2), so exactly 3 − 2 = 1 dimension is crushed (nullity = 1, the normal line).
- **Sanity Check on Derivatives:** On polynomials of degree $\le n$, dim($\mathcal{P}_n$) = $n + 1$. The derivative operator crushes constant polynomials to zero (nullity = 1). Therefore, the range must have dimension exactly ($n + 1$) − 1 = n ($\mathcal{P}_{n-1}$).

Part IV: The Matrix Representation of a Linear Transformation

Chapter 6: The Systematic Construction Recipe ($[T]_{B_V \to B_W}$)

Gilbert Strang famously synthesized the essence of Chapter 8: *"A transformation is geometry without axes; choosing a basis creates the matrix."* Once a basis is chosen for the domain and codomain, any linear transformation between finite-dimensional vector spaces is equivalent to matrix multiplication.

▶ The Systematic 3-Step Recipe for Constructing Matrix $A = [T]^{B_V \to B_W}$

Let V have basis $B_V = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ and let W have basis $B_W = \{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m\}$. The $m \times n$ matrix representation A is computed systematically column by column:

1. **Step 1 (Transform the Basis):** Apply T to the j-th basis vector of the domain: compute $T(\mathbf{v}_j) \in W$.
2. **Step 2 (Expand in Output Basis):** Express $T(\mathbf{v}_j)$ as a unique linear combination of the output basis vectors B_W:

$$T(\mathbf{v}_j) = a_{1j}\,\mathbf{w}_1 + a_{2j}\,\mathbf{w}_2 + \ldots + a_{mj}\,\mathbf{w}_m = \sum_{i=1}^{m} a_{ij}\,\mathbf{w}_i$$

3. **Step 3 (Package into Column *j*):** The expansion coefficients $[a_{1j}, a_{2j}, \ldots, a_{mj}]^T$ become the ***j*-th column** of matrix A:

$$A = [T]_{B_V \to B_W} = \Big| \; [T(\mathbf{v}_1)]_{B_W} \quad [T(\mathbf{v}_2)]_{B_W} \quad \ldots \quad [T(\mathbf{v}_n)]_{B_W} \; \Big| = \begin{vmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \&ddots; & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{vmatrix}$$

The Master Representation Equation: For any arbitrary vector $\mathbf{x} \in V$:

$$[T(\mathbf{x})]_{B_W} = [T]_{B_V \to B_W} \bullet [\mathbf{x}]_{B_V} = A\,[\mathbf{x}]_{B_V}$$

Chapter 7: Function Spaces: Derivative & Integral Operators

One of the most profound realizations in linear algebra is that calculus operations on polynomials are simply matrix multiplications! Let $\mathscr{P}_3$ be the 4-dimensional space of real polynomials of degree ≤ 3, with standard monomial basis: $B = \{1, x, x^2, x^3\}$. Any polynomial $p(x) = c_0 + c_1 x + c_2 x^2 + c_3 x^3$ is uniquely represented by its coordinate vector $[p]_B = [c_0, c_1, c_2, c_3]^T$.

7.1 The Derivative Matrix *D*

Apply the 3-step recipe to the derivative operator $D(p) = \frac{dp}{dx}$:

- $D(1) = 0 = 0(1) + 0(x) + 0(x^2) + 0(x^3)$ &implies; Column 1 = $[0, 0, 0, 0]^T$
- $D(x) = 1 = 1(1) + 0(x) + 0(x^2) + 0(x^3)$ &implies; Column 2 = $[1, 0, 0, 0]^T$
- $D(x^2) = 2x = 0(1) + 2(x) + 0(x^2) + 0(x^3)$ &implies; Column 3 = $[0, 2, 0, 0]^T$
- $D(x^3) = 3x^2 = 0(1) + 0(x) + 3(x^2) + 0(x^3)$ &implies; Column 4 = $[0, 0, 3, 0]^T$

Assembling columns yields the 4 × 4 matrix representation of differentiation:

$$D = \begin{vmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{vmatrix}$$

7.2 The Integration Matrix *S*

Consider the integration operator from $\mathscr{P}_2$ to $\mathscr{P}_3$: $S(p) = \int_0^x p(t)\,dt$. Input basis $B_2 = \{1, x, x^2\}$, output basis $B_3 = \{1, x, x^2, x^3\}$:

- $S(1) = x = [0, 1, 0, 0]^T$
- $S(x) = \frac{1}{2}x^2 = [0, 0, 1/2, 0]^T$
- $S(x^2) = \frac{1}{3}x^3 = [0, 0, 0, 1/3]^T$

$$S = \begin{vmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1/3 \end{vmatrix}$$

The Fundamental Theorem of Calculus in Matrix Form: Multiplying $D \bullet S$:

$$D_{4\times4}\,S_{4\times3} = \begin{vmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{vmatrix} \begin{vmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1/3 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{vmatrix}$$

Restricted to $\mathscr{P}_2$, $D \bullet S = I_{3\times3}$! Differentiation is the left-inverse of integration:

$\frac{d}{dx}\int_0^x p(t)dt = p(x)$.

⚡ FAST-TRACK 3: The Nilpotent Superdiagonal Derivative Matrix

Structural Invariants of the Differentiation Matrix *D*:

1. **Strictly Upper Triangular:** All main diagonal entries are strictly 0. Therefore:

$$\det(D) = 0, \quad \operatorname{tr}(D) = 0, \quad \text{All eigenvalues } \lambda_1 = \lambda_2 = \ldots = \lambda_n = 0$$

2. **Nilpotency Index = dim(*V*):** The matrix is *nilpotent*. Powers of *D* shift entries one step higher along the superdiagonals: D^2 computes the 2nd derivative, D^3 computes the 3rd, and $D^4 = \mathbf{0}$. Every polynomial of degree ≤ 3 vanishes after 4 derivatives!
3. **Kernel & Rank:** The first column is all zeros, so $[1, 0, 0, 0]^T$ (constant polynomials) spans ker(*D*). Nullity = 1, Rank = 3.

Part V: Change of Basis & Coordinate Transformations

Chapter 8: Coordinate Vector Relativity & The Change of Basis Matrix M

A physical vector **x** exists independently of any chosen coordinate system. However, its numerical address — the list of coordinate numbers $[\mathbf{x}]_B$ — depends entirely on the reference basis vectors chosen to measure it.

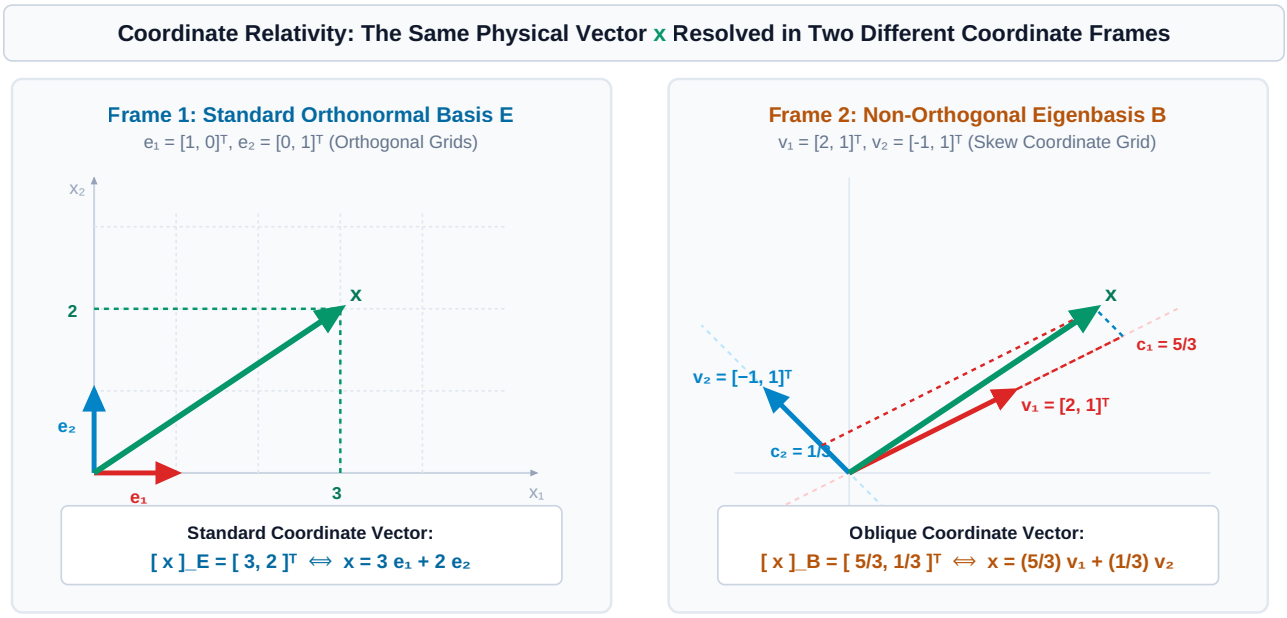


Figure 5.4: Coordinate Relativity: The identical physical vector **x** resolved in standard Cartesian coordinates vs oblique eigenbasis coordinates.

▶ Definition 8.1: The Change of Basis Matrix M

Let $B = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ be the "old" basis, and let $B' = \{\mathbf{v}'_1, \ldots, \mathbf{v}'_n\}$ be the "new" basis. The **Change of Basis Matrix** M (from B' to B) transforms new coordinates into old coordinates:

$$[\mathbf{x}]_B = M\,[\mathbf{x}]_{B'}, \quad \text{where} \quad M = \begin{vmatrix} [\mathbf{v}'_1]_B & [\mathbf{v}'_2]_B & \cdots & [\mathbf{v}'_n]_B \end{vmatrix}$$

Key Rule: The columns of M are the *new basis vectors* expressed in terms of the *old basis*.
To transform in reverse (from old coordinates to new coordinates), multiply by the inverse:

$$[\mathbf{x}]_{B'} = M^{-1}\,[\mathbf{x}]_B$$

Special Case: When the old basis B is the standard basis E, M is simply the matrix formed by writing the new vectors $\mathbf{v}'_j$ directly as columns!

Chapter 9: Similarity Transformations ($B = M^{-1}AM$) & The Commutative Diagram

Suppose a linear transformation $T: V \to V$ is represented by matrix A in basis B. What is its new matrix representation B when the basis is changed to B'?

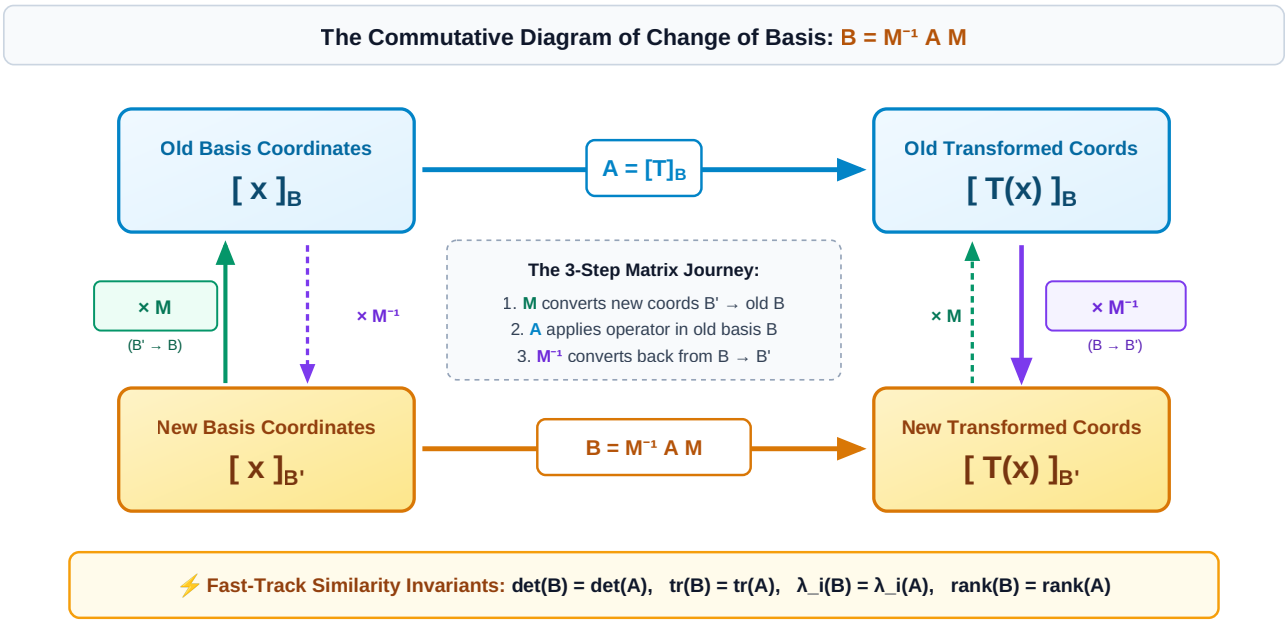


Figure 5.3: The Commutative Diagram of Change of Basis, linking coordinates under A and B via transition matrices M and M^{-1}.

▶ Theorem 9.1: The Similarity Transformation Rule

Let $A = [T]_B$ be the matrix of T in basis B, and let M be the change of basis matrix from B' to B. Then the matrix $B = [T]_{B'}$ of T in the new basis B' is:

$$B = M^{-1} A M$$

Two matrices A and B satisfying this relation are called **similar matrices** (denoted $B \sim A$). They represent the *exact same underlying linear transformation* viewed through different coordinate glasses.

🔍 Derivation via the 3-Step Commutative Walk

Start with the new coordinates of an input vector: $[\mathbf{x}]_{B'}$.

1. **Step 1 (Convert to Old):** Multiply by M to express the vector in the old basis: $[\mathbf{x}]_B = M\,[\mathbf{x}]_{B'}$.
2. **Step 2 (Apply Old Transformation):** Multiply by A to apply operator T in the old basis: $[T(\mathbf{x})]_B = A\,[\mathbf{x}]_B = A\,M\,[\mathbf{x}]_{B'}$.
3. **Step 3 (Return to New):** Multiply by M^{-1} to translate the transformed result back to new coordinates: $[T(\mathbf{x})]_{B'} = M^{-1}\,[T(\mathbf{x})]_B = M^{-1}\,A\,M\,[\mathbf{x}]_{B'}$.

Comparing with the defining equation $[T(\mathbf{x})]_{B'} = B\,[\mathbf{x}]_{B'}$ immediately reveals:

$$B = M^{-1}\,A\,M \quad \blacksquare$$

⚡ FAST-TRACK 4: Basis Change Direction Mnemonic: The M−1AM Sandwich

Never mix up whether M or M^{-1} goes on the right:

1. **The Sandwich Mnemonic:** Read the matrix product from right to left as a physical round trip:

   New Coordinates $\longrightarrow_{\times M}$ **Old Coordinates** $\longrightarrow_{\times A}$ **Transformed Old** $\longrightarrow_{\times M^{-1}}$ **Transformed New**

   M must meet the new input coordinates first to convert them to old, so ***M* is on the right**. M^{-1} brings the answer back, so ***M*$^{-1}$ is on the left**!
2. **Spectral Invariants Under Similarity:** Every geometric property of the operator is invariant under similarity transformations:

   $$\det(M^{-1}AM) = \det(A), \quad \operatorname{tr}(M^{-1}AM) = \operatorname{tr}(A), \quad \lambda_i(B) = \lambda_i(A), \quad \operatorname{rank}(B) = \operatorname{rank}(A)$$

   If your computed B has a different determinant or trace than A, you made an arithmetic sign error in M^{-1}!

Part VI: The Search for a Good Basis

Chapter 10: Eigenbases & Diagonalization ($\Lambda = X^{-1}AX$)

The central quest of linear algebra is to choose coordinate axes so that complicated, coupled physical systems decouple into simple, independent 1D scalar operations.

▶ Theorem 10.1: The Optimal Basis for Operators (The Eigenbasis)

Suppose an $n \times n$ matrix A possesses n linearly independent eigenvectors $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ with corresponding eigenvalues $\lambda_1, \dots, \lambda_n$ (so $A\mathbf{x}_j = \lambda_j\mathbf{x}_j$). If we choose the eigenvectors as our new basis $B' = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$:

1. The change of basis matrix is the eigenvector matrix $X = [\mathbf{x}_1 \dots \mathbf{x}_n]$.
2. Under the 3-step recipe, $T(\mathbf{x}_j) = \lambda_j\mathbf{x}_j = 0\mathbf{x}_1 + \dots + \lambda_j\mathbf{x}_j + \dots + 0\mathbf{x}_n$.
3. Therefore, the matrix representation collapses into a purely diagonal matrix Λ:

$$\Lambda = X^{-1} A X = \begin{vmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \&ddots; & \vdots \\ 0 & 0 & \dots & \lambda_n \end{vmatrix}$$

In the eigenbasis, the n-variable coupled matrix-vector equation $\mathbf{y} = A\mathbf{x}$ uncouples completely into n independent scalar multiplications: $y'_j = \lambda_j x'_j$.

Chapter 11: Singular Vector Bases for General Rectangular Mappings ($\Sigma = U^{-1}AV$)

What if the matrix is $m \times n$ rectangular, or what if A fails to have a full set of eigenvectors? For a mapping between two different spaces $T: \mathbb{R}^n \to \mathbb{R}^m$, we are free to choose **two different bases**: an input basis V for $\mathbb{R}^n$ and an output basis U for $\mathbb{R}^m$.

▶ Theorem 11.1: Singular Value Decomposition as Optimal Basis Selection

For any real $m \times n$ matrix A of rank r, there exist two orthonormal bases:

- Orthonormal input basis $V = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ for $\mathbb{R}^n$ (eigenvectors of $A^\mathsf{T}A$)
- Orthonormal output basis $U = \{\mathbf{u}_1, \ldots, \mathbf{u}_m\}$ for $\mathbb{R}^m$ (eigenvectors of AA^T)

such that each basis vector maps directly to its partner scaled by the singular value σ_j:

$$A\,\mathbf{v}_j = \sigma_j\,\mathbf{u}_j \quad (\text{for } j \le r), \qquad A\,\mathbf{v}_j = \mathbf{0} \quad (\text{for } j > r)$$

In these optimal orthonormal bases, the matrix representation becomes the diagonal matrix Σ:

$$\Sigma = U^{-1} A\, V = U^\mathsf{T} A\, V = \begin{vmatrix} \sigma_1 & \cdots & 0 & \cdots & 0 \\ \vdots & \&ddots; & \vdots & \vdots & \vdots \\ 0 & \cdots & \sigma_r & \cdots & 0 \\ 0 & \cdots & 0 & \cdots & 0 \end{vmatrix}$$

Geometric meaning: V rotates the input unit sphere to align with principal axes, Σ scales along those axes by σ_j, and U rotates the resulting ellipsoid into its final orientation in $\mathbb{R}^m$.

Part VII: Rigorous Worked Problems (Step-by-Step with Verification)

Worked Problem 1: Composite 2D Geometric Transformation Matrix

Geometric Transformations in $\mathbb{R}^2$

PROBLEM STATEMENT

In $\mathbb{R}^2$, find the standard 2 × 2 matrix C representing a reflection across the line $y = 2x$, immediately followed by a counterclockwise rotation through an angle $\theta = 45°$. Verify your result on test vectors and confirm its determinant and orthogonality properties.

STEP 1: CONSTRUCT THE REFLECTION MATRIX H

The line $y = 2x \Leftrightarrow 2x - y = 0$ has normal direction $\mathbf{m} = [2, -1]^T$. Normalizing $\mathbf{m}$ yields the unit normal: $\mathbf{n} = \frac{1}{\sqrt{5}}[2, -1]^T$. Using the Householder reflection formula $H = I - 2\mathbf{n}\mathbf{n}^T$:

$$H = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} - 2 \bullet \frac{1}{5} \begin{vmatrix} 2 \\ -1 \end{vmatrix} \begin{vmatrix} 2 & -1 \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} - \frac{2}{5} \begin{vmatrix} 4 & -2 \\ -2 & 1 \end{vmatrix} = \frac{1}{5} \begin{vmatrix} -3 & 4 \\ 4 & 3 \end{vmatrix}$$

Check on line direction: $\mathbf{d} = [1, 2]^T$ gives $H\mathbf{d} = \frac{1}{5}[-3 + 8, 4 + 6]^T = [1, 2]^T$ (fixed!).

Check on normal: $\mathbf{n} = [2, -1]^T$ gives $H\mathbf{n} = \frac{1}{5}[-6 - 4, 8 - 3]^T = [-2, 1]^T = -\mathbf{n}$ (flipped!).

STEP 2: CONSTRUCT THE ROTATION MATRIX R

For $\theta = 45°$, $\cos 45° = \sin 45° = \frac{\sqrt{2}}{2}$:

$$R = \frac{\sqrt{2}}{2} \begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix}$$

STEP 3: MULTIPLY TO FORM THE COMPOSITE MATRIX C = R • H

Since reflection occurs first, H acts first (on the right): $C = R \bullet H$:

$$C = \left| \frac{\sqrt{2}}{2} \begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix} \right| \bullet \left| \frac{1}{5} \begin{vmatrix} -3 & 4 \\ 4 & 3 \end{vmatrix} \right| = \frac{\sqrt{2}}{10} \begin{vmatrix} (1)(-3) + (-1)(4) & (1)(4) + (-1)(3) \\ (1)(-3) + (1)(4) & (1)(4) + (1)(3) \end{vmatrix} = \frac{\sqrt{2}}{10} \begin{vmatrix} -7 & 1 \\ 1 & 7 \end{vmatrix}$$

Physical & Algebraic Sanity Verification:

• **Determinant:** $\det(C) = \det(R) \bullet \det(H) = (+1)(-1) = -1$. From the explicit matrix: $\det(C) = \left(\frac{2}{100}\right)[(-7)(7) - (1)(1)] = \frac{2}{100}(-50) = -1$. Matches exactly!

• **Orthogonality:** $C^TC = \frac{2}{100} \begin{vmatrix} -7 & 1 \\ 1 & 7 \end{vmatrix} \begin{vmatrix} -7 & 1 \\ 1 & 7 \end{vmatrix} = \frac{2}{100} \begin{vmatrix} 49+1 & -7+7 \\ -7+7 & 1+49 \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = I$. Rigid isometry preserved!

Worked Problem 2: Differential Operator on Polynomial Space $\mathscr{P}_3$ & ODE Kernel

Function Space Operators

PROBLEM STATEMENT

Consider the linear differential operator T: $\mathscr{P}_3 \to \mathscr{P}_3$ defined by: $T(p) = (x + 1)\,p'(x) - 2\,p(x)$, where $\mathscr{P}_3$ has standard basis $B = \{1, x, x^2, x^3\}$.

(a) Construct the 4×4 matrix representation $A = [T]_B$.

(b) Find the kernel $\ker(T)$, its dimension (nullity), and the corresponding polynomial solution to $T(p) = 0$.

(c) Find the range of T and verify the Dimension Theorem.

STEP 1: EVALUATE T ON EACH BASIS POLYNOMIAL

- $p_0 = 1$ &implies; $p'_0 = 0$ &implies; $T(1) = (x+1)(0) - 2(1) = -2 =$ **$-2(1)$**
- $p_1 = x$ &implies; $p'_1 = 1$ &implies; $T(x) = (x+1)(1) - 2x = 1 - x =$ **$1(1) - 1(x)$**
- $p_2 = x^2$ &implies; $p'_2 = 2x$ &implies; $T(x^2) = (x+1)(2x) - 2x^2 = 2x =$ **$0(1) + 2(x)$**
- $p_3 = x^3$ &implies; $p'_3 = 3x^2$ &implies; $T(x^3) = (x+1)(3x^2) - 2x^3 = x^3 + 3x^2 =$ **$0(1) + 0(x) + 3(x^2) + 1(x^3)$**

STEP 2: ASSEMBLE MATRIX A = $[T]_B$

$$A = \begin{vmatrix} -2 & 1 & 0 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

Matrix A is upper triangular! Its eigenvalues are the diagonal entries: $\lambda \in \{-2, -1, 0, 1\}$. Because $\lambda = 0$ is an eigenvalue, $\ker(T)$ is non-trivial.

STEP 3: SOLVE FOR KERNEL KER(T) AND ODE SOLUTION

Solve $A\mathbf{c} = \mathbf{0}$ for coordinate vector $\mathbf{c} = [c_0, c_1, c_2, c_3]^T$:

- Row 4: $1 \cdot c_3 = 0$ &implies; $c_3 = 0$
- Row 3: $0c_2 + 3c_3 = 0$ &implies; $0 = 0$. Thus c_2 is a **free variable**! Choose $c_2 = 1$.
- Row 2: $-c_1 + 2c_2 = 0$ &implies; $c_1 = 2c_2 = 2$
- Row 1: $-2c_0 + c_1 = 0$ &implies; $2c_0 = c_1 = 2$ &implies; $c_0 = 1$

The nullspace vector is $\mathbf{c} = [1, 2, 1, 0]^T$. The corresponding polynomial is:

$$p(x) = 1 + 2x + x^2 = (x + 1)^2$$

Direct ODE Check: $T((x+1)^2) = (x+1) \cdot 2(x+1) - 2(x+1)^2 = 2(x+1)^2 - 2(x+1)^2 = 0$! Perfectly exact!

Dimension Theorem Verification:

- nullity = $\dim(\ker T) = 1$ (spanned by $(x+1)^2$).
- Columns 1, 2, and 4 are pivot columns (linearly independent), so rank = $\dim(\text{range } T) = 3$.
- Dimension Theorem: $\dim(\ker T) + \dim(\text{range } T) = 1 + 3 = 4 = \dim(\mathscr{P}_3)$. Q.E.D.

Worked Problem 3: Change of Basis Matrix & Oblique Coordinate Vector Resolution

Coordinate Transformations

PROBLEM STATEMENT

In $\mathbb{R}^2$, let $E = \{\mathbf{e}_1, \mathbf{e}_2\}$ be the standard basis, and consider the alternative basis: $B = \{\mathbf{v}_1, \mathbf{v}_2\}$ with $\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$.

(a) Compute the change of basis matrix M from B to E, and its inverse M^{-1} from E to B.

(b) Find the coordinate vector $[\mathbf{x}]_B$ for $\mathbf{x} = \begin{bmatrix} 7 \\ 2 \end{bmatrix}$.

(c) Verify algebraically by synthesizing the linear combination.

STEP 1: CONSTRUCT M AND M^{-1}

Because E is the standard basis, the columns of M are simply the vectors $\mathbf{v}_1$ and $\mathbf{v}_2$:

$$M = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}, \quad \det(M) = (2)(1) - (-1)(1) = 2 + 1 = 3 \neq 0$$

Using the 2×2 fast inversion shortcut (swap diagonal, negate off-diagonal, divide by det):

$$M^{-1} = \frac{1}{3} \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix}$$

STEP 2: TRANSFORM COORDINATES FROM STANDARD TO BASIS B

$$[\mathbf{x}]_B = M^{-1}[\mathbf{x}]_E = \frac{1}{3}\begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 7 \\ 2 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} (1)(7) + (1)(2) \\ (-1)(7) + (2)(2) \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 9 \\ -3 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

Linear Combination Verification:

Reconstitute vector $\mathbf{x}$ from its oblique coordinates $c_1 = 3$, $c_2 = -1$:

$3\mathbf{v}_1 + (-1)\mathbf{v}_2 = 3\begin{bmatrix} 2 \\ 1 \end{bmatrix} - 1\begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 - (-1) \\ 3 - 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 2 \end{bmatrix} = \mathbf{x}$. The coordinates $[3, -1]^T$ represent vector $\mathbf{x}$ perfectly!

Worked Problem 4: Similarity Transformation & Operator Diagonalization

Similarity & Eigenbases

PROBLEM STATEMENT

In the standard basis of $\mathbb{R}^2$, a linear transformation is given by $A = \begin{vmatrix} 4 & 2 \\ 1 & 3 \end{vmatrix}$.

(a) Find the eigenvalues and eigenvectors of A.

(b) Form the eigenbasis B' and its transition matrix X.

(c) Compute $B = X^{-1}AX$ step-by-step and verify that B is diagonal.

(d) Verify determinant and trace invariance under similarity.

STEP 1: COMPUTE EIGENVALUES & EIGENVECTORS

$$\det(A - \lambda I) = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 5)(\lambda - 2) = 0$$

Eigenvalues are $\lambda_1 = 5$ and $\lambda_2 = 2$.

- For $\lambda_1 = 5$: $(A - 5I)\mathbf{x} = \begin{vmatrix} -1 & 2 \\ 1 & -2 \end{vmatrix} \begin{vmatrix} x_1 \\ x_2 \end{vmatrix} = \mathbf{0}$ &implies; $\mathbf{x}_1 = \begin{vmatrix} 2 \\ 1 \end{vmatrix}$
- For $\lambda_2 = 2$: $(A - 2I)\mathbf{x} = \begin{vmatrix} 2 & 2 \\ 1 & 1 \end{vmatrix} \begin{vmatrix} x_1 \\ x_2 \end{vmatrix} = \mathbf{0}$ &implies; $\mathbf{x}_2 = \begin{vmatrix} -1 \\ 1 \end{vmatrix}$

STEP 2: CONSTRUCT TRANSITION MATRIX X AND X^{-1}

$$X = \begin{vmatrix} 2 & -1 \\ 1 & 1 \end{vmatrix}, \quad \det(X) = 3, \quad X^{-1} = \frac{1}{3} \begin{vmatrix} 1 & 1 \\ -1 & 2 \end{vmatrix}$$

STEP 3: STEP-BY-STEP MULTIPLICATION OF X^{-1} A X

First, evaluate the right-hand product AX:

$$AX = \begin{vmatrix} 4 & 2 \\ 1 & 3 \end{vmatrix} \begin{vmatrix} 2 & -1 \\ 1 & 1 \end{vmatrix} = \begin{vmatrix} 8 + 2 & -4 + 2 \\ 2 + 3 & -1 + 3 \end{vmatrix} = \begin{vmatrix} 10 & -2 \\ 5 & 2 \end{vmatrix} = \begin{vmatrix} 5\mathbf{x}_1 & 2\mathbf{x}_2 \end{vmatrix}$$

Now multiply by X^{-1} on the left:

$$B = X^{-1}(AX) = \frac{1}{3} \begin{vmatrix} 1 & 1 \\ -1 & 2 \end{vmatrix} \begin{vmatrix} 10 & -2 \\ 5 & 2 \end{vmatrix} = \frac{1}{3} \begin{vmatrix} 10 + 5 & -2 + 2 \\ -10 + 10 & 2 + 4 \end{vmatrix} = \frac{1}{3} \begin{vmatrix} 15 & 0 \\ 0 & 6 \end{vmatrix} = \begin{vmatrix} 5 & 0 \\ 0 & 2 \end{vmatrix} = \Lambda$$

Similarity Invariants Audit:

- **Trace Invariance:** $\text{tr}(A) = 4 + 3 = 7$. $\text{tr}(\Lambda) = 5 + 2 = 7$. Matches!
- **Determinant Invariance:** $\det(A) = (4)(3) - (2)(1) = 10$. $\det(\Lambda) = (5)(2) = 10$. Matches!
- **Conclusion:** In the eigenbasis, the off-diagonal coupling terms vanish completely, producing pure decoupled scaling along $\mathbf{x}_1$ and $\mathbf{x}_2$.

Worked Problem 5: Rectangular Linear Map Between Different Vector Spaces

Cross-Space Representations & Interpolation

PROBLEM STATEMENT

Define the evaluation-derivative operator $T: \mathscr{P}_2 \to \mathbb{R}^3$ by:

$$T(p) = \begin{vmatrix} p(0) \\ p(1) \\ p'(1) \end{vmatrix}$$

Let $\mathscr{P}_2$ have monomial basis $B = \{1, x, x^2\}$ and let $\mathbb{R}^3$ have standard basis E.

(a) Construct the 3 × 3 matrix representation $A = [T]_{B \to E}$.

(b) Prove that T is an isomorphism by inverting matrix A.

(c) Use A^{-1} to find the unique quadratic polynomial $p(x)$ satisfying $p(0) = 2$, $p(1) = 5$, and $p'(1) = 4$.

STEP 1: EVALUATE T ON BASIS POLYNOMIALS

- $p_0 = 1$: $p(0) = 1$, $p(1) = 1$, $p'(1) = 0$ &implies; Column 1 = $[1, 1, 0]^T$
- $p_1 = x$: $p(0) = 0$, $p(1) = 1$, $p'(1) = 1$ &implies; Column 2 = $[0, 1, 1]^T$
- $p_2 = x^2$: $p(0) = 0$, $p(1) = 1$, $p'(1) = 2$ &implies; Column 3 = $[0, 1, 2]^T$

$$A = \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 2 \end{vmatrix}$$

STEP 2: INVERT MATRIX A

Expanding det(A) along Row 1: det(A) = 1 • [(1)(2) − (1)(1)] = 1 ≠ 0. Since det ≠ 0, T is an isomorphism!

Computing the inverse matrix A^{-1} via cofactor transpose:

$$A^{-1} = \begin{vmatrix} 1 & 0 & 0 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{vmatrix}$$

Check: $A \bullet A^{-1} =$

$$\begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 2 \end{vmatrix} \begin{vmatrix} 1 & 0 & 0 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = I.$$

STEP 3: SOLVE THE HERMITE INTERPOLATION PROBLEM

We are given the target output vector $\mathbf{y} = [2, 5, 4]^T$. The coordinate vector $[p]_B$ is:

$$[p]_B = A^{-1}\begin{vmatrix}2\\5\\4\end{vmatrix} = \begin{vmatrix}1 & 0 & 0\\-2 & 2 & -1\\1 & -1 & 1\end{vmatrix}\begin{vmatrix}2\\5\\4\end{vmatrix} = \begin{vmatrix}(1)(2)+0+0\\(-2)(2)+(2)(5)+(-1)(4)\\(1)(2)+(-1)(5)+(1)(4)\end{vmatrix} = \begin{vmatrix}2\\-4+10-4\\2-5+4\end{vmatrix} = \begin{vmatrix}2\\2\\1\end{vmatrix}$$

The unique interpolating polynomial is: $\boldsymbol{p(x) = 2 + 2x + x^2}$.

Direct Polynomial Verification:

• $p(0) = 2 + 0 + 0 =$ **2**. Matches!

• $p(1) = 2 + 2(1) + (1)^2 = 2 + 2 + 1 =$ **5**. Matches!

• $p'(x) = 2 + 2x$ &implies; $p'(1) = 2 + 2(1) =$ **4**. Matches!

Algebraic interpolation solved purely through matrix inversion!

Part VIII: Comprehensive Master Reference & Cheat Sheet

Chapter 12: Master Reference Summary Table

Concept / Operator	Governing Formula / Matrix	Determinant	Trace / Invariants	Key Property & Geometric Meaning
2D Rotation R_θ	$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$	+1	$2\cos\theta$	Orthogonal, preserves Euclidean lengths, angles & orientation. $R^{-1} = R^T$.
2D Reflection $H_{2\alpha}$	$\begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{bmatrix}$	−1	0	Involutory ($H^2 = I$), symmetric, reverses handedness. Eigenvalues: ±1.
Orthogonal Projection P	$\mathbf{a}\mathbf{a}^T = \begin{bmatrix} a_1^2 & a_1a_2 \\ a_1a_2 & a_2^2 \end{bmatrix}$	0	1	Idempotent ($P^2 = P$), symmetric. Eigenvalues: 1, 0. Crushes normal line.
Horizontal Shear $S_x(k)$	$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$	+1	2	Area-preserving deformation. Only 1 independent eigenvector $[1, 0]^T$. Defective.
Non-Uniform Scaling D	$\begin{bmatrix} s_1 & 0 \\ 0 & s_2 \end{bmatrix}$	$s_1 s_2$	$s_1 + s_2$	Scales principal axes independently. Eigenvalues: s_1, s_2.
Dimension Theorem	$\dim(\ker T) + \dim(\text{range } T) = \dim V$	—	Nullity + Rank = n	Conservation of input degrees of freedom across any linear transformation.
Representation Recipe	$A = [[T(\mathbf{v}_1)]_{B_W} \dots [T(\mathbf{v}_n)]_{B_W}]$	—	$[T(\mathbf{x})] = A[\mathbf{x}]$	Columns are images of domain basis vectors expanded in codomain basis.
Change of Basis	$[\mathbf{x}]_B = M\,[\mathbf{x}]_{B'}$ $\quad (M = [[\mathbf{v}'_j]_B])$	$\det(M) \neq 0$	$[\mathbf{x}]_{B'} = M^{-1}\,[\mathbf{x}]_B$	Columns of M are new basis vectors written in the old coordinate frame.
Similarity Transformation	$B = M^{-1} A M$	$\det(B) = \det(A)$	$\text{tr}(B) = \text{tr}(A)$	Same physical operator expressed in two different coordinate bases. λ_i invariant.
Diagonalization	$\Lambda = X^{-1} A X = \text{diag}(\lambda_1, \dots, \lambda_n)$	$\prod \lambda_i$	$\sum \lambda_i$	Decouples multi-variable operator into n independent 1D scalar stretches.
Singular Value Decomp.	$\Sigma = U^T A V = \text{diag}(\sigma_1, \dots, \sigma_r)$	—	$A\mathbf{v}_j = \sigma_j \mathbf{u}_j$	Optimal orthonormal input and output bases for general rectangular transformations.

⚡ FAST-TRACK MASTER RECAP: 4 Golden Rules for Exams and Research

1. **Rule 1 (The Column Rule):** To find the matrix of any transformation, don't write general algebra. Just ask: *"Where does [1,0]^T land? Where does [0,1]^T land?"* Those destinations are your columns. Done in 5 seconds.
2. **Rule 2 (The Dimension Conservation Law):** Always verify nullity + rank = dim(Domain). Lost dimensions + Surviving dimensions = Starting dimensions.
3. **Rule 3 (The M^{-1}AM Sandwich Rule):** New coords enter on the right, so M must meet them first to convert to old: AM. M^{-1} brings the result back on the left: $B = M^{-1}AM$.
4. **Rule 4 (The Similarity Invariant Guard):** Before submitting an answer for $B = M^{-1}AM$, instantly sum the diagonals (trace) and check the determinant. If tr(B) ≠ tr(A) or det(B) ≠ det(A), stop and fix your signs immediately!